

The Electron Alphabet, the Cayley–Hypercube Family, and a Threefold Characterisation of $n = 4$

Towards a Combinatorial Language Layer for the PDL-V Programme

PDL Document DL03

Cédric Laubscher

Independent Researcher, Switzerland

cedriclaubscher.ch

July 2026

Abstract

DL01 and DL02 establish that thresholds n_{vie} , n_{cons} for the emergence of life and consciousness in the PDL framework exist and are necessarily distinct, without fixing their numerical value [1, 2]. This document reports three unconditional, computationally verified results obtained while probing what a combinatorial *alphabet* carried by admissible closures would need to look like, as a candidate route towards giving that threshold operational content. First, the set of V_4 -orbits of $\text{Coh}(K_4)$ — already known to number five, by the theorem of D60 [3] — is shown here to form a linear code of minimum Hamming distance three, with the striking additional property that every minimal-distance pair changes orbit: the electron’s admissible configurations are not merely five in number, they are maximally non-redundant. Second, K_4 and C_4 are shown to be, respectively, the dense and sparse Cayley graphs on the same group $\mathbb{Z}_2 \times \mathbb{Z}_2$, embedding the electron in an infinite family (the hypercubes Q_k) whose richness under the analogous $\mathbb{Z}_2^k$ construction grows super-exponentially ($5 \rightarrow 30 \rightarrow 2288$ for $k = 2, 3, 4$). Third, we exhibit a threefold, independently-derived characterisation of $n = 4$ — the triangle-count theorem of DL01/DL02, the spin-resolution theorem of D66 [4], and a new orbit-regularity criterion proved here — and show by exact polynomial division that all three share the linear factor $(n - 4)$. None of these results establishes a numerical value for n_{vie} or n_{cons} ; they are reported as a first, verified layer of combinatorial structure on which such a determination might eventually be built.

Contents

1	Scope and Relation to DL01/DL02	3
2	The Electron Alphabet as a Linear Code	3
2.1	The Five V_4 -Orbits (Restated from D60)	3
2.2	$\text{Coh}(K_4)$ is a Linear Code	3
2.3	Minimal Pairs are Maximally Informative	4
2.4	Generalisation to the Hypercube Family	4
3	The Cayley–Hypercube Family	4
3.1	K_4 and C_4 are Cayley Graphs on the Same Group	4
3.2	The Hypercube Growth Law	5
4	A Threefold Characterisation of $n = 4$	5
4.1	Three Independent Derivations	5
4.2	Exact Algebraic Convergence	6
4.3	Robustness of the Orbit-Regularity Criterion	6
5	Relation to the PDL-V Programme	7

1 Scope and Relation to DL01/DL02

DL01 proves that n_{vie} and n_{cons} exist as finite thresholds and are necessarily distinct; DL02 refines this with the fraction $f(n)$ of coherent configurations among all configurations of K_n , showing it decreases double-exponentially, and identifies $\mathcal{R}_{\text{active}}^1$ as true for all but the homogeneous configuration [1, 2]. Neither document fixes a numerical value for either threshold.

This document does not attempt to fix that value either. It reports a different kind of progress: an exploration of what a combinatorial *alphabet* — a discrete, non-redundant, generatively rich set of distinguishable configurations carried by admissible closures — would look like if one existed, undertaken independently of, but directly relevant to, the DL01/DL02 programme. The results below are unconditional theorems or exhaustively verified computations; none is speculative. Where an interpretive link to the language-of-life question is drawn, it is explicitly marked as such and separated from the underlying mathematics.

2 The Electron Alphabet as a Linear Code

2.1 The Five V_4 -Orbits (Restated from D60)

D60 proves that C1-admissibility on $\text{Coh}(K_4)$ is equivalent to invariance under V_4 , that V_4 acts regularly on the generic orbit of $\text{Coh}(K_4)$, and fixes the remaining four configurations pointwise [3]. The resulting orbit structure is

$$\text{Coh}(K_4)/V_4 = \{1, 1, 1, 1, 4\}, \quad (1)$$

five orbits in total: the identity configuration and the three perfect matchings M_1, M_2, M_3 of K_4 (later identified with the three $SU(3)_c$ colour axes [5]) as singleton orbits, and one generic orbit of size four. This result is not re-derived here; it is the starting point for the new results of this section.

2.2 $\text{Coh}(K_4)$ is a Linear Code

Theorem 2.1

Identify $\text{Coh}(K_4) \subset \{\pm 1\}^6$ (six edges of K_4) with a binary code via $\pm 1 \leftrightarrow \{0, 1\}$. Then $\text{Coh}(K_4)$ is closed under coordinatewise multiplication: for all $c_1, c_2 \in \text{Coh}(K_4)$, $c_1 \cdot c_2 \in \text{Coh}(K_4)$, where $(c_1 \cdot c_2)_e = (c_1)_e(c_2)_e$ for each edge e .

Computational Verification 2.1

Verified by exhaustive check over all $\binom{8}{2}$ ordered pairs of the eight elements of $\text{Coh}(K_4)$: every product lies in $\text{Coh}(K_4)$. Equivalently, $\text{Coh}(K_4)$ is a subgroup of $\{\pm 1\}^6$ under the pointwise product, with identity element the all-plus configuration.

Theorem 2.2

The Hamming weight distribution of $\text{Coh}(K_4)$ (number of -1 entries per configuration) is $\{0, 3, 3, 3, 3, 4, 4, 4\}$. Consequently, by the standard identity for linear codes (minimum distance equals minimum non-zero weight), the minimum Hamming distance of $\text{Coh}(K_4)$ is

$$d_{\min}(\text{Coh}(K_4)) = 3. \quad (2)$$

A code of length six, eight codewords, and minimum distance three saturates the single-error-correcting regime: by the Hamming bound, any single-bit corruption of a codeword is guaranteed

to leave $\text{Coh}(K_4)$, and hence be detectable; the code additionally identifies the corrupted bit uniquely, since $d_{\min} = 3 \geq 2t + 1$ for $t = 1$.

2.3 Minimal Pairs are Maximally Informative

Theorem 2.3

Let a *minimal pair* be two elements of $\text{Coh}(K_4)$ at Hamming distance $d_{\min} = 3$. There are sixteen such pairs (of $\binom{8}{2} = 28$ total pairs; the remaining twelve are at distance four). Every one of the sixteen minimal pairs lies in *different* V_4 -orbits.

Computational Verification 2.2

Verified exhaustively: for all sixteen pairs (c_1, c_2) with $d(c_1, c_2) = 3$, the canonical V_4 -orbit representatives (computed as $\min_{g \in V_4} g \cdot c$ under a fixed total order) differ. Zero of the sixteen pairs share an orbit.

Theorem 2.3 is a stronger and independent statement from Theorem 2.2: it says that the code's closest pairs, without exception, are also orbit-distinct under the very group (V_4) that C1-admissibility forces. There is no redundancy at the finest resolution the code offers: every minimal perturbation of an admissible configuration changes which of the five distinguishable classes it belongs to.

2.4 Generalisation to the Hypercube Family

Theorem 2.4

The same two properties (linearity, and 100% orbit-change at minimal Hamming distance $d_{\min} = 3$) hold for $\text{Coh}(Q_3)$ under the translation group $\mathbb{Z}_2^3$ (128 codewords, verified exhaustively over all 512 minimal pairs).

This shows Theorems 2.1–2.3 are not an artefact of K_4 's small size, but a property of (at least the first two members of) the family constructed in Section 3.

3 The Cayley–Hypercube Family

3.1 K_4 and C_4 are Cayley Graphs on the Same Group

Theorem 3.1

Let $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$. Then:

$$K_4 \cong \text{Cay}(\mathbb{Z}_2 \times \mathbb{Z}_2, S_{\text{dense}}), \quad S_{\text{dense}} = \{(1, 0), (0, 1), (1, 1)\} \text{ (all non-zero elements),} \quad (3)$$

$$C_4 \cong \text{Cay}(\mathbb{Z}_2 \times \mathbb{Z}_2, S_{\text{sparse}}), \quad S_{\text{sparse}} = \{(1, 0), (0, 1)\} \text{ (unit-weight generators).} \quad (4)$$

Computational Verification 3.1

Verified by direct graph isomorphism check ($\text{Cay}(\mathbb{Z}_2 \times \mathbb{Z}_2, S_{\text{dense}}) \cong K_4$ and $\text{Cay}(\mathbb{Z}_2 \times \mathbb{Z}_2, S_{\text{sparse}}) \cong C_4$, both confirmed).

Theorem 3.1 identifies K_4 and C_4 as two densities of the same underlying algebraic object, not as unrelated small graphs. This motivates the natural generalisation to higher k .

3.2 The Hypercube Growth Law

For $Q_k = \text{Cay}(\mathbb{Z}_2^k, S_{\text{sparse}})$ (the standard k -cube), define L_k as the number of orbits of $\text{Coh}(Q_k)$ under the coordinate-flip group $\mathbb{Z}_2^k$ (translation by each generator).

Table 1: Growth of L_k for the first three non-trivial hypercubes.

k	$ \text{Coh}(Q_k) = 2^{2^k-1}$	L_k	Ratio L_k/L_{k-1}
2 ($= C_4$)	8	5	—
3	128	30	6.0
4	32768	2288	76.3

Computational Verification 3.2

$L_2 = 5$ and $L_3 = 30$ verified by exhaustive orbit enumeration; $L_4 = 2288$ verified by exhaustive orbit enumeration over all 32768 coherent configurations under the sixteen-element group $\mathbb{Z}_2^4$, with orbit-size distribution {1:16, 4:140, 8:240, 16:1892}.

The ratio in Table 1 is itself accelerating ($6.0 \rightarrow 76.3$), not merely growing: L_k diverges super-exponentially in k . Combined with Section 2.4, this identifies a family — the Cayley graphs on $\mathbb{Z}_2^k$ at hypercube density — in which every tested member is simultaneously combinatorially rich (Table 1) and, for the two smallest members, a maximally non-redundant linear code (Theorems 2.1–2.3, 2.4).

Open Problem 3.1

Prove or refute Theorems 2.1–2.3 for Q_4 and, if a pattern is confirmed, for general Q_k . The exhaustive method used for Q_2, Q_3 is computationally intractable at $k = 4$ without exploiting the algebraic (linear-code) structure directly; a proof exploiting Theorem 2.1’s group structure, rather than brute-force enumeration, is the natural next step.

4 A Threefold Characterisation of $n = 4$

4.1 Three Independent Derivations

Three separate constructions, from three separate parts of the corpus and this document, single out $n = 4$ as the unique solution to an equation on K_n .

Theorem 4.1

(DL01/DL02.) K_4 is the unique fixed point, among complete graphs, of an operator S whose fixed-point condition reduces to $C(n, 3) = n(n-1)(n-2)/6 = 4$ [1, 2].

Theorem 4.2

(D66.) $n = 4$ is the unique value $n > 1$ for which the resolved spin exponent $2n - 2$ equals the edge count $\binom{n}{2}$, i.e. $2n - 2 = \binom{n}{2}$, verified computationally for $n = 2, \dots, 11$ [4].

Theorem 4.3

(This document.) $n = 4$ is the unique value, among $n = 2, \dots, 28$, for which $\text{Coh}(K_n)$ possesses a non-trivial orbit (under the full symmetric group S_n , via the bipartition parametrisation of Harary’s theorem) of size exactly four — the order required for a Klein four-group to act regularly upon it.

Computational Verification 4.1

Orbit sizes under S_n (bipartition classes $k = 0, \dots, \lfloor n/2 \rfloor$, size $\binom{n}{k}$ for $k \neq n - k$, $\binom{n}{k}/2$ otherwise) were enumerated exhaustively for $n = 2, \dots, 28$. Exactly one non-trivial match to size four was found: $n = 4$, $k = 1$ ($\binom{4}{1} = 4$). No other (n, k) pair in this range yields size four.

4.2 Exact Algebraic Convergence

Theorem 4.4

The polynomials underlying Theorems 4.1 and 4.2 share the linear factor $(n - 4)$ exactly:

$$n(n - 1)(n - 2) - 24 = (n - 4)(n^2 + n + 6), \quad (5)$$

$$n(n - 1) - 2(2n - 2) = (n - 4)(n - 1). \quad (6)$$

Verified by exact polynomial division (zero remainder in both cases). The two remaining factors are of different character: $n^2 + n + 6$ has no real root (discriminant -23), while $n - 1$ has the real root $n = 1$, excluded from D66’s domain by the stated condition $n > 1$. Theorem 4.3 is not a polynomial identity of this type: it is a statement about which (n, k) pairs realise a given binomial coefficient exactly, and it likewise resolves uniquely to $n = 4$, with no residual factor at all.

4.3 Robustness of the Orbit-Regularity Criterion

The construction underlying Theorem 4.3 additionally distinguishes between the two groups of order four.

Theorem 4.5

At $n = 4$, the Klein four-group V_4 acts regularly on the size-four orbit and fixes the remaining four configurations pointwise (this is D60’s theorem, Section 2). The cyclic group $\mathbb{Z}_4 \leq S_4$ acts regularly on the same orbit but does *not* fix the remaining four configurations pointwise.

Computational Verification 4.2

Verified directly: the four-cycle generator of $\mathbb{Z}_4 \leq S_4$, applied to each of the four singleton-orbit configurations under V_4 , does not return each to itself.

Theorem 4.5 shows that the regularity half of Theorem 4.3’s criterion (an order-four group can act transitively on a four-element set) is a fact about group order alone, by Lagrange’s theorem, and holds for any order-four group; but the fixing half is specific to the Klein structure of V_4 and fails for the cyclic alternative. This is consistent with, and gives a partial structural reason for, why D60’s theorem selects V_4 rather than an arbitrary order-four subgroup of S_4 .

Open Problem 4.1

Construct an explicit bijection between the three mechanisms of Theorems 4.1–4.3, establishing that they are three readings of a single underlying constraint rather than three independent coincidences sharing a common factor. The shared factor $(n - 4)$ (Theorem 4.4) is verified exactly but not yet explained from a common first principle.

5 Relation to the PDL-V Programme

The results of this document do not determine n_{vie} or n_{cons} . What they establish is a verified layer of structure directly upstream of that question: the electron’s alphabet is not an arbitrary five-element set but a maximally efficient error-correcting code (Section 2); this code sits inside an infinite, super-exponentially growing family (Section 3); and the base case of that family, $n = 4$, is forced with unusual redundancy — by three structurally unrelated arguments converging on the same algebraic root (Section 4). A threshold-search programme building on DL01/DL02 could take the growth law of Table 1, together with a signal-transmission fidelity criterion of the kind investigated separately, as a candidate operational definition of when a closure hierarchy has become “rich enough”; this document supplies the alphabet and growth law half of such a definition, not the threshold itself.

References

- [1] Cédric Laubscher. From axioms to life: The pdl-v programme. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.20132166>.
- [2] Cédric Laubscher. Life and consciousness thresholds in the projective dynamic logo framework. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.20132228>.
- [3] Cédric Laubscher. $G_{\text{eff}} = S_4/V_4 \cong S_3$: The H_{SU2} theorem in the projective dynamic logo framework. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.20639684>.
- [4] Cédric Laubscher. Search for the c5 metric candidate: Logical geometry, $(a) \wedge (b)$, and the parity obstruction of K_4 in the projective dynamic logo framework. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.21351177>.
- [5] Cédric Laubscher. $SU(3)_c$ in the carrier space w of the projective dynamic logo framework. *Zenodo*, 2026. <https://doi.org/10.5281/zenodo.20629282>.